

## EBS, EFS and VPC

### Class no 15

#### How to make a snapshot for EBS volume?

Consider we have a volume named: vol-0e67333a18e71b278. Snapshot is the backup for the volume.

To perform that,

**Steps no:1** Click on the volume → actions → create a snapshot

Volumes (1/1) Info

Recycle Bin Actions Create volume

Modify volume  
Create snapshot  
Create snapshot lifecycle policy  
Delete volume  
Attach volume  
Detach volume  
Force detach volume  
Manage auto-enabled I/O

| Name                  | Volume ID | Type   | Size | IOPS | Throughput | Snapshot ID | Source volume ID |
|-----------------------|-----------|--------|------|------|------------|-------------|------------------|
| vol-0e67333a18e71b278 | gp3       | 20 GiB | 3000 | 125  | -          | -           | -                |

**Step no:2** Name it → create snapshot.

Create snapshot Info

Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.

Source volume

Volume ID: vol-0e67333a18e71b278

Availability Zone: eun1-az2 (eu-north-1b)

Snapshot details

Description: Add a description for your snapshot

Mysnapshot

255 characters maximum.

Encryption: Not encrypted

Tags

No tags associated with the resource.

Add new tag

Cancel Create snapshot

We have to wait until it becomes completed.

Snapshots (1) Info

Recycle Bin Actions Create snapshot

| Name                   | Snapshot ID | Full snapshot size | Volume size | Description | Storage tier | Snapshot status            | Started | Progress |
|------------------------|-------------|--------------------|-------------|-------------|--------------|----------------------------|---------|----------|
| snap-0ccccb79c98f17d83 | 72.5 MiB    | 20 GiB             | Mysnapshot  | Standard    | Completed    | 2026/03/31 20:02 GMT+5:... | 100%    |          |

After that we can create a volume from that snapshot using the snapshot Id.

EC2 > Snapshots > snap-0ccccb79c98f17d83

Details

Snapshot ID: snap-0ccccb79c98f17d83

Owner: 314383175192

Description: Mysnapshot

Full snapshot size: 72.5 MiB

Started: Tue Mar 31 2026 20:02:36 GMT+0530 (India Standard Time)

Progress: 100%

Product codes: -

## How to create a volume from a snapshot and reduce the size?

**Step no:1** Go to EC2 instance → scroll down → volume → create volume.

Inside volume choose the configuration size as 20 gb reduced from 100gb.

Choose snapshot id: paste it (snap-0ccccb79c98f17d83). Create volume.

Volume settings

Volume type | Info

General Purpose SSD (gp3)

Size (GiB) | Info

20

Min: 1 GiB, Max: 65536 GiB

IOPS | Info

3000

Min: 3000 IOPS, Max: 80000 IOPS

Throughput (MiB/s) | Info

125

Min: 125 MiB, Max: 2000 MiB, Baseline: 125 MiB/s

Availability Zone | Info

eu-north-1a (eu-north-1a)

Snapshot ID - optional | Info

Specify custom snapshotid...

Fast snapshot restore | Info

Not enabled for selected snapshot

Custom Snapshot ID

Specify custom snapshotid

Snapshot ID

snap-0ccccb79c98f17d83

Cancel Save

## How to create an AMI?(Amazon Machine Image)

**Step no:1** AMI is the backup for our EC2 instance along with the volume. Launch an instance.

Select it → actions → image and templates → create image

Instances (1/2) | Info

Last updated 12 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets

Choose filter set Find Instance by attribute or tag (case-sensitive)

| Name              | Instance ID         | Instance state | Instance type | Status check    | Alarm status  | Availability Zone | Public IPv4 DNS      |
|-------------------|---------------------|----------------|---------------|-----------------|---------------|-------------------|----------------------|
| My linux insta... | i-0b22922c2d19d1f39 | Running        | t3.micro      | 3/3 checks pass | View alarms + | eu-north-1b       | ec2-13-53-168-146... |
| My linux Insta... | i-05384bc783349ea8b | Running        | t3.micro      | 3/3 checks pass | View alarms + | eu-north-1a       | ec2-13-53-168-146... |

Create image

Create template from instance

Launch more like this

Image and templates

Storage

Monitor and troubleshoot

**Step no:2** Name it → choose: tag image and snapshots together (it includes volume backup) → create.

Image details

Instance ID

i-0b22922c2d19d1f39 (My linux instance 2)

Image name

MyInstanceSnapshot

Image description - optional

MyInstanceSnapshot

Maximum 255 characters

☒ Reboot instance

When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

| Storage type | Device     | Snapshot                      | Size | Volume type                   | IOPS | Throughput | Delete on termination                      | Encrypted                       |
|--------------|------------|-------------------------------|------|-------------------------------|------|------------|--------------------------------------------|---------------------------------|
| EBS          | /dev/xv... | Create new snapshot from v... | 8    | EBS General Purpose SSD - ... | 3000 |            | <input checked="" type="checkbox"/> Enable | <input type="checkbox"/> Enable |

Add volume

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

Tags - optional

Tag image and snapshots together

Tag the image and the snapshots with the same tag.

Tag image and snapshots separately

Tag the image and the snapshots with different tags.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Create image

## **Elastic File System:**

- **Elastic File System (EFS)** is a **serverless file storage service** provided by AWS.
- It automatically scales storage capacity up or down as files are added or removed.
- Designed for use with **Linux-based workloads** via the **NFS (Network File System) protocol**.

### **Key Features**

#### **1. Serverless Service**

- No need to provision or manage storage servers.
- AWS handles scaling and availability automatically.

#### **2. Dependent Service – EC2**

- Commonly mounted on **Amazon EC2 instances**.
- Multiple EC2 instances can access the same EFS simultaneously.

#### **3. Multi-AZ Support**

- EFS is **region-wide** and spans multiple **Availability Zones (AZs)**.
- Ensures high availability and durability.

#### **4. Shared File System**

- Acts like a shared drive.
- Multiple servers can read/write to the same files at the same time.

#### **5. NFS Protocol**

- Uses standard **NFSv4** protocol.
- Compatible with Linux machines.

#### **6. Drive Link Concept**

- You can host the **same website/application** across multiple servers by linking them to the same EFS drive. Ensures consistency of data across servers.

### **Use Cases**

- **Web Hosting:** Store website files in EFS so multiple EC2 servers can serve the same content.
- **Big Data & Analytics:** Shared storage for distributed processing.
- **Content Management:** Centralized storage for media files accessed by multiple applications.
- **Backup & Restore:** Scalable storage for backups.

### **Benefits**

- **Scalable:** Grows/shrinks automatically with usage.
- **Highly Available:** Data replicated across multiple AZs.
- **Shared Access:** Multiple EC2 instances can access simultaneously.
- **Simple Integration:** Mountable via NFS on Linux systems.

In short: **EFS = Shared, scalable, serverless file system for EC2 and Linux workloads, available across multiple AZs.**

**(Important Note:** It supports concurrent access from thousands of EC2 instances access multiple Availability Zones.)

### Hands on: To create a EFS shared volume for EC2 Linux Instances:

**Step no:1** Create two Linux instances: My Linux instance 1 and My Linux instance 2.

Note that The EFS file system is only available for Linux. Hence it is not applicable for windows.

Instances (2) [Info](#)

Last updated less than a minute ago

Saved filter sets  
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

| <input type="checkbox"/> | Name <a href="#">✎</a> ▼ | Instance ID         | Instance state ▼                            | Instance type ▼ | Status check      | Alarm status                  |
|--------------------------|--------------------------|---------------------|---------------------------------------------|-----------------|-------------------|-------------------------------|
| <input type="checkbox"/> | My linux insta...        | i-0b22922c2d19d1f39 | Running <a href="#">ⓘ</a> <a href="#">🔍</a> | t3.micro        | 3/3 checks passed | <a href="#">View alarms +</a> |
| <input type="checkbox"/> | My linux Insta...        | i-05384bc783349ea8b | Running <a href="#">ⓘ</a> <a href="#">🔍</a> | t3.micro        | 3/3 checks passed | <a href="#">View alarms +</a> |

While creating choose network settings → edit → add security group rule → NFS → 2049. Do it for both the instance and add custom and source anywhere and don't disturb the existing security group rule ssh.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) [Remove](#)

Type [Info](#)

ssh ▼

Source type [Info](#)

Anywhere ▼

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source [Info](#)

Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - optional [Info](#)

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 2049, 0.0.0.0/0) [Remove](#)

Type [Info](#)

NFS ▼

Source type [Info](#)

Custom ▼

Protocol [Info](#)

TCP

Port range [Info](#)

2049

Source [Info](#)

Add CIDR, prefix list or security group

0.0.0.0/0 ✕

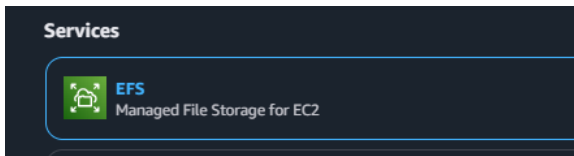
Description - optional [Info](#)

e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Add security group rule

## Step no:2 Search EFS.



Create File system → name it as 'myFileSystem' → keep others default → Create file system.

**Create file system**

Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

**Name (optional)**

Name your file system.

myFileSystem

Name can include letters, numbers, and -./:/ symbols, up to 256 characters.

**Virtual Private Cloud (VPC)**

Choose the VPC where you want EC2 instances to connect to your file system.

vpc-08d23a3e5bc324ead

**Recommended settings**

Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#).

| Setting                                    | Value                       | Editable after creation |
|--------------------------------------------|-----------------------------|-------------------------|
| Throughput mode <a href="#">Learn more</a> | Elastic                     | Yes                     |
| Transition into Infrequent Access (IA)     | 30 day(s) since last access | Yes                     |
| Transition into Archive                    | 90 day(s) since last access | Yes                     |
| Transition into Standard                   | None                        | Yes                     |
| Automatic backups                          | Enabled                     | Yes                     |
| Encryption                                 | Enabled                     | No                      |

Cancel Customize **Create file system**

**Step no:3** Now click on the EFS that we have created, under that we have option called network → copy the security group (anyone and ensure everything is default).

Eg. sg-0d22bad617206e07

**myFileSystem (fs-039c395982f53bf30)** Delete Attach

**General** Edit

Amazon resource name (ARN)

arn:aws:elasticfilesystem:eu-north-1:314383175192:file-system/fs-039c395982f53bf30

**Performance mode**

General Purpose

**Throughput mode**

Elastic

**Lifecycle management**

Transition into Infrequent Access (IA): 30 day(s) since last access  
Transition into Archive: 90 day(s) since last access  
Transition into Standard: None

**Availability zone**

Regional

**Automatic backups**

Enabled

**Encrypted**

68cf7ae3-1a80-4706-8ee2-87758722a63c (aws/elasticfilesystem)

**File system state**

Available

**DNS name**

fs-039c395982f53bf30.efs.eu-north-1.amazonaws.com

**Replication overwrite protection**

Enabled

**Network** Manage

| Availability zone (AZ-ID) | Mount target ID        | Subnet ID                | VPC ID                                | Mount target state | IPv4 address  | IPv6 address | Network interface ID  | Security groups                      |
|---------------------------|------------------------|--------------------------|---------------------------------------|--------------------|---------------|--------------|-----------------------|--------------------------------------|
| eu-north-1a (eu1-az1)     | fsmt-0c89daaca1b807c2f | subnet-0dc90cfff302b4e0a | <a href="#">vpc-08d23a3e5bc324ead</a> | Available          | 172.31.20.119 | N/A          | eni-05e73f7698d126e12 | <b>sg-0d22bad617206e07 (default)</b> |
| eu-north-1b (eu1-az2)     | fsmt-054f46f891f194c23 | subnet-0532105e641bc86bf | <a href="#">vpc-08d23a3e5bc324ead</a> | Available          | 172.31.33.26  | N/A          | eni-0e03b4ba48960317d | sg-0d22bad617206e07 (default)        |
| eu-north-1c (eu1-az3)     | fsmt-06d6c82497f8a8445 | subnet-0b82d32bcc749d0fa | <a href="#">vpc-08d23a3e5bc324ead</a> | Available          | 172.31.4.243  | N/A          | eni-0bcb80efb21a24caa | sg-0d22bad617206e07 (default)        |

Now we need to make the NFS is in the security group.

Go to instances → security groups → Enter the copied id in the search filter (sg-0d22bad617206e0) → Open.

**sg-0d22bad617206e077 - default** Actions ▾

**Details**

|                                |                                           |                                            |                                 |
|--------------------------------|-------------------------------------------|--------------------------------------------|---------------------------------|
| Security group name<br>default | Security group ID<br>sg-0d22bad617206e077 | Description<br>default VPC security group  | VPC ID<br>vpc-08d23a3e5bc324ead |
| Owner<br>314383175192          | Inbound rules count<br>1 Permission entry | Outbound rules count<br>1 Permission entry |                                 |

**Inbound rules** | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

**Inbound rules (1)** Manage tags Edit inbound rules

| Name | Security group rule ID | IP version | Type        | Protocol | Port range | Source                 | Description |
|------|------------------------|------------|-------------|----------|------------|------------------------|-------------|
| -    | sgr-0b8f3bfb81a291657  | -          | All traffic | All      | All        | sg-0d22bad617206e07... | -           |

Edit inbound rules → Add rule → NFS → 0.0.0.0/0 → save rules.

This is to allow NFS in the inbound rules, even though it allows all traffic we need to mention the NFS to avoid any disturbance.

**Edit inbound rules** Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

**Inbound rules** Info

| Security group rule ID | Type        | Protocol | Port range | Source      | Description - optional |        |
|------------------------|-------------|----------|------------|-------------|------------------------|--------|
| sgr-0b8f3bfb81a291657  | All traffic | All      | All        | Custom      |                        | Delete |
| -                      | NFS         | TCP      | 2049       | Anywhere... | 0.0.0.0/0              | Delete |

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

**Step no:4** Connect our instance 1. Follow the commands,

Switch user: **sudo su**

Create a directory: **mkdir efsnew**

```
Last login: Wed Apr 1 06:08:45 2026 from 13.48.4.202
[ec2-user@ip-172-31-47-179 ~]$ sudo su
[root@ip-172-31-47-179 ec2-user]# mkdir efsnew
```

Now in our created EFS click the attach → mount through DNS → copy the code for NFS client.

**Attach**

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS ☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-039c395982f53bf30:/ efs
```

Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrns=2,noresvport fs-039c395982f53bf30.efs.eu-north-1.amazonaws.com:/ efs
```

See our user guide for more information. [Learn more](#)

Close

Now in the EC2 console paste the code and ensure that instead of 'efs' at the end we have the respective directory name 'efsnew'.

```
Last login: Wed Apr 1 06:08:45 2026 from 13.48.4.202
[ec2-user@ip-172-31-47-179 ~]$ sudo su
[root@ip-172-31-47-179 ec2-user]# mkdir efsnew
[root@ip-172-31-47-179 ec2-user]# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-039c395982f53bf30.efs.eu-north-1.amazonaws.com:/ efsnew
```

Now do the same for the EC2 instance 2. Name the directory as **efsold1**.

```
Last login: Wed Apr 1 06:15:56 2026 from 13.48.4.203
[ec2-user@ip-172-31-32-121 ~]$ sudo su
[root@ip-172-31-32-121 ec2-user]# mkdir efsold1
[root@ip-172-31-32-121 ec2-user]# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-039c395982f53bf30.efs.eu-north-1.amazonaws.com:/ efsold1
```

**Step no:5** Get inside of the directory: efsold1. It is inside the EC2 2 instance.

Command: **cd efsold1/**

Create a file named happy123file.txt using **touch filename**

```
[root@ip-172-31-32-121 ec2-user]# cd efsold1/
[root@ip-172-31-32-121 efsold1]# touch happy123file.txt
[root@ip-172-31-32-121 efsold1]#
```

→ EC2 2

Now come to our EC2 → 1 instance and open the **efsnew** directory. List the files we can see the **happy123file.txt** under this also.

```
[root@ip-172-31-47-179 home]# cd /home/ec2-user
[root@ip-172-31-47-179 ec2-user]# cd efsnew/
[root@ip-172-31-47-179 efsnew]# ls
happy123file.txt
[root@ip-172-31-47-179 efsnew]#
```

→ EC2 1

**(Note:** As EFS is a shared file system common for multiple EC2 on different AZ, whatever the changes on the EC2 → 2 it will be reflected on the EC2 → 1 and vice versa.)

## Virtual Private Network (VPC)

Amazon VPC (Virtual Private Cloud) is the backbone of AWS networking, allowing you to create a logically isolated environment with full control over IP ranges, subnets, routing, and security. Its key components—subnets, route tables, gateways, and firewalls—work together to define how resources communicate internally and externally.

### VPC and Its Components

#### 1. VPC Basics

- A Virtual Private Cloud (VPC) is a logically isolated section of the AWS cloud.
- You define IP address ranges (CIDR blocks) for your VPC.
- Provides complete control over networking: subnets, routing, gateways, and security.

#### 2. Subnets

- Public Subnet: Connected to an Internet Gateway; resources here can access the internet directly.
  - Common use: Load balancers, bastion hosts, NAT gateways.
- Private Subnet: No direct internet route; resources only have private IPs.
  - Common use: Databases, application servers.

#### 3. Route Tables

- Define how traffic flows within the VPC.
- Each subnet must be associated with a route table.
- Example: Public subnet routes → Internet Gateway; Private subnet routes → NAT Gateway.

#### 4. Internet Gateway (IGW)

- A horizontally scaled, redundant gateway that **connects VPC resources to the internet**.
- Required for public subnets.

#### 5. NAT Gateway / NAT Instance

- Allows instances in **private subnets to access the internet securely** (e.g., for updates).
- Prevents inbound internet traffic to private resources.

#### 6. Security Groups

- Act as instance-level firewalls.
- Control inbound and outbound traffic at the EC2 instance level.
- Stateful: return traffic is automatically allowed.

#### 7. Network ACLs (Access Control Lists)

- Subnet-level firewalls.
- Stateless: rules must be defined for both inbound and outbound traffic.
- Useful for fine-grained subnet security.

#### 8. VPC Peering

- Connects two VPCs for resource sharing.
- Traffic between peered VPCs uses private IPs, not the public internet.



### Reference Table:

| Component        | Purpose                               | Scope          | Example Use Case           |
|------------------|---------------------------------------|----------------|----------------------------|
| Subnets          | Divide VPC into public/private zones  | VPC-level      | Hosting EC2, DBs           |
| Route Tables     | Define traffic paths                  | Subnet-level   | Internet access via IGW    |
| Internet Gateway | Internet connectivity                 | VPC-level      | Public-facing apps         |
| NAT Gateway      | Outbound internet for private subnets | Subnet-level   | DB patching                |
| Security Groups  | Instance firewall                     | Instance-level | Allow SSH, block others    |
| Network ACLs     | Subnet firewall                       | Subnet-level   | Block suspicious IP ranges |
| VPC Peering      | Connect VPCs                          | Cross-VPC      | Multi-region apps          |

### CIDR Calculation: (Classless Interdomain Range)

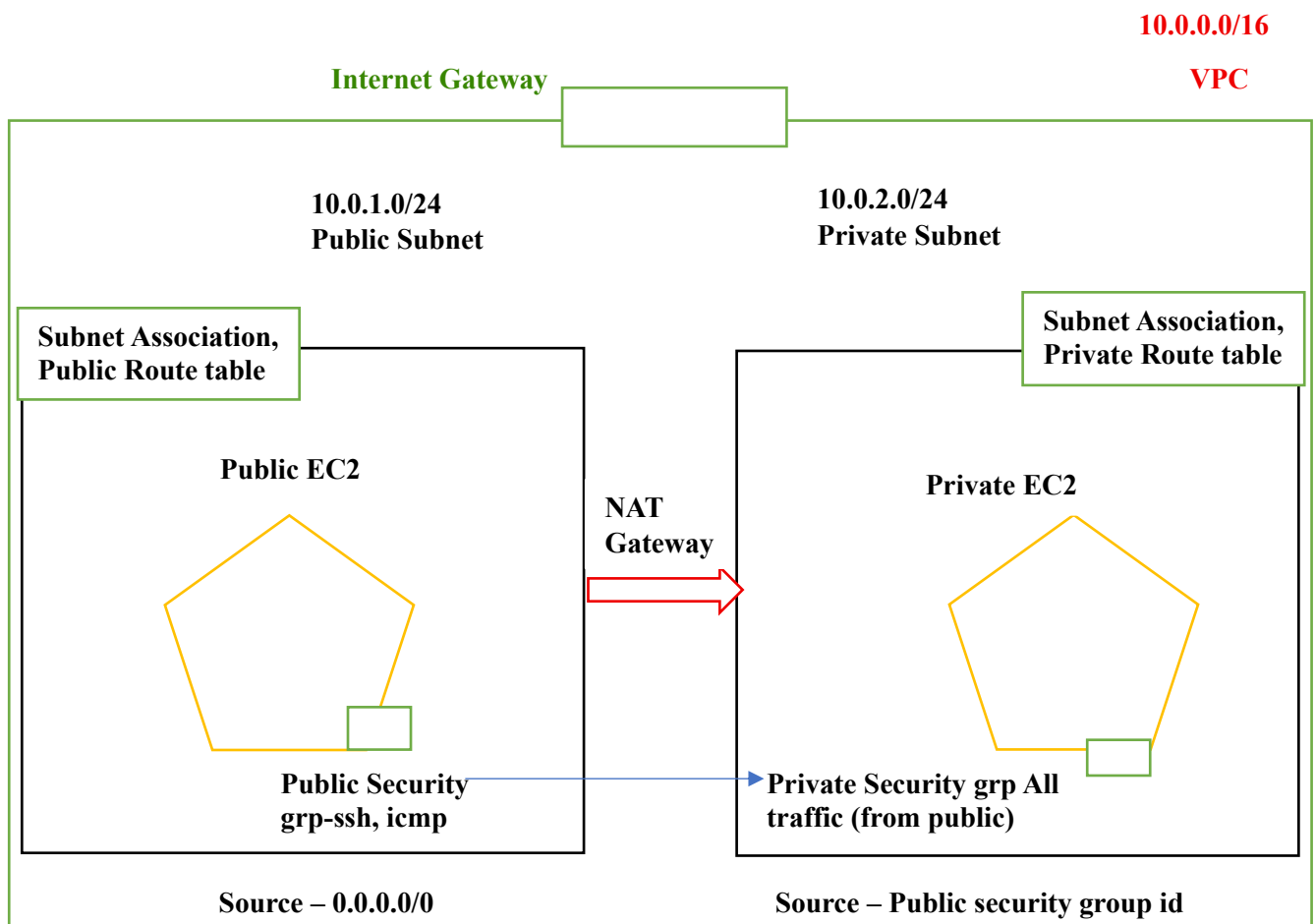
IPv4 → 32 bits (0.0.0.0 – 255.255.255.255) , IPv6 → 64 bits

Example calculation: 10.0.0.0/16 (IPv4)

$32 - 16 = 16$

$2^{16} = 65536$  (Count of Ip address we can have in this range.)

### VPC Framework Architecture:



## Hands on: Creating a VPC

### Step no:1 Search VPC → Create VPC.

Resource to create: **VPC only** → Name it: **my-vpc-01**

CIDR range: **10.0.0.0/16** → Create VPC.

**VPC settings**

**Resources to create** [Info](#)  
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

**Name tag - optional**  
Creates a tag with a key of 'Name' and a value that you specify.

my-vpc-01

**IPv4 CIDR block** [Info](#)  
☒ IPv4 CIDR manual input  
☐ IPAM-allocated IPv4 CIDR block

**IPv4 CIDR**  
10.0.0.0/16

CIDR block size must be between /16 and /28.

**IPv6 CIDR block** [Info](#)  
☒ No IPv6 CIDR block  
☐ IPAM-allocated IPv6 CIDR block  
☐ Amazon-provided IPv6 CIDR block  
☐ IPv6 CIDR owned by me

**Tenancy** [Info](#)  
Default

**VPC encryption control (\$)** [Info](#)  
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply [L2](#).

☒ None ☐ Monitor mode ☐ Enforce mode

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key** **Value - optional**

Q Name X Q my-vpc-01 X Remove tag

Add tag

You can add 49 more tags

Cancel Preview code **Create VPC**

### Step no:2 Create two Subnets.

Choose our VPC → Name it: Public Subnet → 10.0.0.0/16 → 10.0.1.0/24. Add new Subnet

Create subnet

**VPC**

**VPC ID**  
Create subnets in this VPC.  
vpc-0f675cd313f60565f (my-vpc-01)

**Associated VPC CIDRs**

**IPv4 CIDRs**  
10.0.0.0/16

**Subnet settings**  
Specify the CIDR blocks and Availability Zone for the subnet.

**Subnet 1 of 1**

**Subnet name**  
Create a tag with a key of 'Name' and a value that you specify.  
Public-Subnet

**Availability Zone** [Info](#)  
Choose the zone in which your subnet will reside, or let Amazon choose one for you.  
Europe (Stockholm) / eu-north-1a

**IPv4 VPC CIDR block** [Info](#)  
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.  
10.0.0.0/16

**IPv4 subnet CIDR block**  
10.0.1.0/24 256 IPs

**Tags - optional**  
Remove

Add new subnet

Cancel **Create subnet**

Add new Subnet.

Private Subnet → 10.0.0.0/16 → 10.0.2.0/24.Create Subnet.

Subnet 2 of 2

**Subnet name**  
Create a tag with a key of 'Name' and a value that you specify.  
Private-Subnet

**Availability Zone** [Info](#)  
Choose the zone in which your subnet will reside, or let Amazon choose one for you.  
Europe (Stockholm) / eu-north-1a

**IPv4 VPC CIDR block** [Info](#)  
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.  
10.0.0.0/16

**IPv4 subnet CIDR block**  
10.0.2.0/24 256 IPs

▼ **Tags - optional**

**Key** **Value - optional**

Q Name X Q Private-Subnet X Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel Create subnet

**Step no:3** Create two route tables.

### Public Route Table:

Name: Public-RT → VPC: my-vpc-01(Enter the tag, The ID will be populated)→ Create Route table.

**Create route table** [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

**Route table settings**

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.  
Public-RT

**VPC**  
The VPC to use for this route table.  
vpc-0f675cd313f60565f (my-vpc-01)

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key** **Value - optional**

Q Name X Q Public-RT X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

### Private Route Table:

Name: Private-RT → VPC: my-vpc-01(Enter the tag, The ID will be populated)→ Create Route table.

**Create route table** [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

**Route table settings**

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.  
Private-RT

**VPC**  
The VPC to use for this route table.  
vpc-0f675cd313f60565f (my-vpc-01)

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key** **Value - optional**

Q Name X Q Private-RT X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

**Step no:4** Create Internet Gateway. Name it: my-internet-gateway → create internet gateway.

### Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

#### Internet gateway settings

##### Name tag

Creates a tag with a key of 'Name' and a value that you specify.

my-internet-gateway

#### Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

##### Key

Q Name

##### Value - optional

Q my-internet-gateway

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

Once create → choose it → actions → attach to VPC.

### Internet gateways (1/2) [Info](#)

Find internet gateways by attribute or tag

|                                     | Name                | Internet gateway ID   | State    | VPC ID                | Owner        |
|-------------------------------------|---------------------|-----------------------|----------|-----------------------|--------------|
| <input type="checkbox"/>            | -                   | igw-01bdc4dff593c53f  | Attached | vpc-08d23a3e5bc324ead | 314383175192 |
| <input checked="" type="checkbox"/> | my-internet-gateway | igw-0a94735a436fa9e36 | Detached | -                     | 314383175192 |

Actions

View details

Attach to VPC

Detach from VPC

Manage tags

Delete internet gateway

Choose: my-vpc-01 → Attach internet gateway.

### Attach to VPC (igw-0a94735a436fa9e36) [Info](#)

#### VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

##### Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0f675cd313f60565f

AWS Command Line Interface command

Cancel

Attach internet gateway

**Step no:5** Create security group.

#### Public SG:

Name: Public-sg

VPC: my-vpc-01

Inbound rules: Add

Ssh → 0.0.0.0/0

Icmp → 0.0.0.0/0

#### Private SG:

Copy the Public sg - ID

Name: Private-security-group

VPC: my-vpc-01

Inbound rules: Add rules

All traffic: sg-068574f951805ab7e (The ID of the public Security group)

This ensures that only the ip addresses passing the public subnet can access the private subnet.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Public-sg

Name cannot be edited after creation.

Description [Info](#)

Public-sg

VPC [Info](#)

vpc-0f675cd313f60565f (my-vpc-01)

Inbound rules [Info](#)

Type [Info](#)

SSH

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source [Info](#)

Anywh...

0.0.0.0/0

Description - optional [Info](#)

Delete

Type [Info](#)

All ICMP - IPv4

Protocol [Info](#)

ICMP

Port range [Info](#)

All

Source [Info](#)

Anywh...

0.0.0.0/0

Description - optional [Info](#)

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type [Info](#)

All traffic

Protocol [Info](#)

All

Port range [Info](#)

All

Destination [Info](#)

Custom

0.0.0.0/0

Description - optional [Info](#)

Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel

Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Private-sg

Name cannot be edited after creation.

Description [Info](#)

Private-sg

VPC [Info](#)

vpc-0f675cd313f60565f (my-vpc-01)

Inbound rules [Info](#)

Type [Info](#)

All traffic

Protocol [Info](#)

All

Port range [Info](#)

All

Source [Info](#)

Custom

sg-068574f951805ab7e

sg-068574f951805ab7e

Description - optional [Info](#)

Delete

Add rule

Outbound rules [Info](#)

Type [Info](#)

All traffic

Protocol [Info](#)

All

Port range [Info](#)

All

Destination [Info](#)

Custom

0.0.0.0/0

Description - optional [Info](#)

Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel

Create security group

## Step no:6 Public Route table association with the Public Subnet.

Choose Public route table → Subnet Associations → Edit Subnet association.

Route tables (1/4) [Info](#)

Find route tables by attribute or tag

| Name                                          | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                             | Owner ID     |
|-----------------------------------------------|-----------------------|---------------------------|-------------------|------|---------------------------------|--------------|
| -                                             | rtb-04ea76a3eacb73669 | -                         | -                 | Yes  | vpc-08d23a3e5bc324ead           | 314383175192 |
| -                                             | rtb-063128fb66f07485c | -                         | -                 | Yes  | vpc-0f675cd313f60565f   my-v... | 314383175192 |
| <input checked="" type="checkbox"/> Public-RT | rtb-04308c77a8efd9717 | -                         | -                 | No   | vpc-0f675cd313f60565f   my-v... | 314383175192 |
| <input type="checkbox"/> Private-RT           | rtb-0d806c51e40fcee4c | -                         | -                 | No   | vpc-0f675cd313f60565f   my-v... | 314383175192 |

rtb-04308c77a8efd9717 / Public-RT

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (0)

Find subnet association

| Name | Subnet ID | IPv4 CIDR | IPv6 CIDR |
|------|-----------|-----------|-----------|
|------|-----------|-----------|-----------|

[Edit subnet associations](#)

Public Subnet → Save association

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

| Name                                              | Subnet ID                | IPv4 CIDR   | IPv6 CIDR |
|---------------------------------------------------|--------------------------|-------------|-----------|
| <input type="checkbox"/> Private-Subnet           | subnet-0c97b04740f889db8 | 10.0.2.0/24 | -         |
| <input checked="" type="checkbox"/> Public-Subnet | subnet-0bad384feac7e62d2 | 10.0.1.0/24 | -         |

Selected subnets

subnet-0bad384feac7e62d2 / Public-Subnet

[Cancel](#) [Save association](#)

Routing it to Internet gateway to connect with public EC2 via internet through the public route table.

Route → Add route → Internet gateway → my Internet gateway → save.

Route tables > rtb-04308c77a8efd9717 > Edit routes

Edit routes

| Destination | Target           | Status | Propagated | Route Origin     |
|-------------|------------------|--------|------------|------------------|
| 10.0.0.0/16 | local            | Active | No         | CreateRouteTable |
| 0.0.0.0/0   | Internet Gateway |        | No         | CreateRoute      |

[Add route](#)

Use: "igw-0a94735a436fa9e36"

igw-0a94735a436fa9e36 (my-internet-gateway)

[Cancel](#) [Preview](#) [Save changes](#)

## Step no:7 Private Route table association with the Private Subnet.

|                                                |                       |   |   |    |                                 |              |
|------------------------------------------------|-----------------------|---|---|----|---------------------------------|--------------|
| <input checked="" type="checkbox"/> Private-RT | rtb-0d806c51e40fcee4c | - | - | No | vpc-0f675cd313f60565f   my-v... | 314383175192 |
|------------------------------------------------|-----------------------|---|---|----|---------------------------------|--------------|

rtb-0d806c51e40fcee4c / Private-RT

Details | Routes | **Subnet associations** | Edge associations | Route propagation | Tags

Explicit subnet associations (0)

Find subnet association

[Edit subnet associations](#)

Private subnet → Save association.

### Edit subnet associations

Change which subnets are associated with this route table.

**Available subnets (1/2)**

| <input type="checkbox"/>            | Name           | Subnet ID                | IPv4 CIDR   | IPv6 CIDR |
|-------------------------------------|----------------|--------------------------|-------------|-----------|
| <input checked="" type="checkbox"/> | Private-Subnet | subnet-0c97b04740f889db8 | 10.0.2.0/24 | -         |
| <input type="checkbox"/>            | Public-Subnet  | subnet-0bad384feac7e62d2 | 10.0.1.0/24 | -         |

**Selected subnets**  
subnet-0c97b04740f889db8 / Private-Subnet X

Cancel Save associations

As the private routable is to connect with Private EC2, which contains sensitive information. We can not provide direct access via internet. Instead we need to create a NAT gateway through which it can access internet securely.

NAT gateway → create → Name: my-nat-gateway-01 → VPC: my-vpc-01 → create NAT gateway.

Create NAT gateway

**NAT gateway settings**  
**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.  
  
Maximum length is 255 characters.

**Availability mode** Info  
Choose whether to deploy across all zones in the region or restrict to a single availability zone.  

☒ **Regional - new**  
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ **Zonal**  
Provides granular control within a specific availability zone, adhering to subnet level settings.

**VPC**  
Select a VPC in which to create the regional NAT gateway.

**Connectivity type**  
Select a connectivity type for the NAT gateway.  

☒ **Public**

☐ Private

**Method of Elastic IP (EIP) allocation** Info  
Choose how IP addresses are associated with NAT gateways.  

☒ **Automatic**  
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling-- adding AZs automatically allocates EIPs, simplifying management.

☐ **Manual**  
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.  

**Key**

**Value - optional**

You can add 49 more tags.

Cancel Create NAT gateway

Routing it to NAT gateway to connect with private EC2 through the private route table.

Route → Add route → NAT gateway → my-nat-gateway-01 → save.

### Edit routes

| Destination                            | Target      | Status | Propagated | Route Origin     |
|----------------------------------------|-------------|--------|------------|------------------|
| 10.0.0.0/16                            | local       | Active | No         | CreateRouteTable |
| <input type="text" value="0.0.0.0/0"/> | NAT Gateway | -      | No         | CreateRoute      |

Use: "nat-1cdb99d338cc340fa"  
nat-1cdb99d338cc340fa (my-nat-gateway-01)

Cancel Preview Save changes

## Step no:8 Create a Public EC2.

Name: Public Ec2, OS: Amazon Linux, provide keypair.

### Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

#### Name and tags [Info](#)

Name

Public Ec2

[Add additional tags](#)

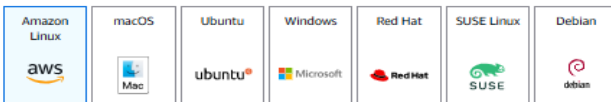
#### ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

**Quick Start**



[Browse more AMIs](#)  
Including AMIs from  
AWS, Marketplace and  
the Community

#### Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI  
ami-0cc38fb663fa09c2 (64-bit (x86), uefi-preferred) / ami-0ae9f10a2843cff29 (64-bit (Arm), uefi)  
Virtualization: hvm    ENA enabled: true    Root device type: ebs

Free tier eligible

## Network Settings:

- My-vpc-01
- Public-subnet
- Auto assign public IP → Enable
- Existing security group → Public sg

Launch instance.

#### ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

batchfrw

[Create new key pair](#)

#### ▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-0f675cd313f60565f (my-vpc-01)  
10.0.0.0/16

[Create new VPC](#)

Subnet | [Info](#)

subnet-0bad384feac7e62d2    Public-Subnet  
VPC: vpc-0f675cd313f60565f    Owner: 314383175192    Availability Zone: eu-north-1a (eu-n1-az1)  
Zone type: Availability Zone    IP addresses available: 251    CIDR: 10.0.1.0/24

[Create new subnet](#)

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups | [Info](#)

Select security groups

[Compare security group rules](#)

Public-sg sg-068574f951805ab7e [X](#)  
VPC: vpc-0f675cd313f60565f

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► **Advanced network configuration**



## Step no:9 Create a Private EC2.

Name: Private Ec2, OS: Amazon Linux, provide keypair.

### Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

#### Name and tags [Info](#)

Name

Private Ec2

[Add additional tags](#)

#### ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

[Quick Start](#)



[Browse more AMIs](#)  
Including AMIs from  
AWS, Marketplace and  
the Community

#### Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI  
ami-0cc38fb663fa09c2 (64-bit (x86), uefi-preferred) / ami-0ae9f10a2843cff29 (64-bit (Arm), uefi)  
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

## Network Settings:

- My-vpc-01
- Private-subnet
- Auto assign public IP → Disable
- Existing security group → Private sg

Launch instance.

#### ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

batchfrw ▼

[Create new key pair](#)

#### ▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-0f675cd313f60565f (my-vpc-01)  
10.0.0.0/16 ▼

[Create new vpc](#)

Subnet | [Info](#)

subnet-0c97b04740f889db8 Private-Subnet ▼  
VPC: vpc-0f675cd313f60565f Owner: 314383175192 Availability Zone: eu-north-1a (eu1-az1)  
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

[Create new subnet](#)

Auto-assign public IP | [Info](#)

Disable ▼

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups | [Info](#)

Select security groups ▼

[Compare security group rules](#)

Private-sg sg-0763f95e192195d71 ✕  
VPC: vpc-0f675cd313f60565f

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► [Advanced network configuration](#)

**Step no: 10** Now connect the public EC2 and check the internet connectivity using: **ping private\_ipaddress\_Public\_EC2**.

Now we can see the internet connectivity is successful in Public Ec2.

```
[ec2-user@ip-10-0-1-43 ~]$ ping 10.0.1.43
PING 10.0.1.43 (10.0.1.43) 56(84) bytes of data.
64 bytes from 10.0.1.43: icmp_seq=1 ttl=127 time=0.026 ms
64 bytes from 10.0.1.43: icmp_seq=2 ttl=127 time=0.034 ms
64 bytes from 10.0.1.43: icmp_seq=3 ttl=127 time=0.033 ms
64 bytes from 10.0.1.43: icmp_seq=4 ttl=127 time=0.028 ms
^C
--- 10.0.1.43 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3143ms
rtt min/avg/max/mdev = 0.026/0.030/0.034/0.003 ms
[ec2-user@ip-10-0-1-43 ~]$
```

**i-051ce309e8e9599bc (Public Ec2)**  
PublicIPs: 51.20.83.241 PrivateIPs: 10.0.1.43

**Step no: 11** To ensure the NAT gateway allows internet to the Private EC2 through public EC2, Ping the private ip of the private EC2 in the Public EC2: **ping private\_ip\_Private\_EC2**.

(Note: Private Ec2 has only one IP address, that is also private)

**i-0e6e10550065aa41b (Private Ec2)**

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

|                                    |                          |
|------------------------------------|--------------------------|
| Instance ID<br>i-0e6e10550065aa41b | Public IPv4 address<br>- |
|------------------------------------|--------------------------|

Private IPv4 addresses  
10.0.2.139

```
ec2-user@ip-10-0-1-43 ~]$ ping 10.0.2.139
PING 10.0.2.139 (10.0.2.139) 56(84) bytes of data.
4 bytes from 10.0.2.139: icmp_seq=1 ttl=127 time=0.427 ms
4 bytes from 10.0.2.139: icmp_seq=2 ttl=127 time=0.220 ms
4 bytes from 10.0.2.139: icmp_seq=3 ttl=127 time=0.157 ms
4 bytes from 10.0.2.139: icmp_seq=4 ttl=127 time=0.177 ms
4 bytes from 10.0.2.139: icmp_seq=5 ttl=127 time=0.153 ms
4 bytes from 10.0.2.139: icmp_seq=6 ttl=127 time=0.157 ms
4 bytes from 10.0.2.139: icmp_seq=7 ttl=127 time=0.160 ms
```

